Datapitch - Finance

Anton Pühringer - 12414573
Maximilian Trentini - 11916686
Michele Fu Wang - 12347518
Julian Bridi - 12301867
Jonathan Gabriel - 12319994

05.11.2026

Agenda

- **Stock Return Analysis & Modelling (Fama-French)**
  - Data, Research Question and Method
  - The Fama-French Three-Factor Model
  - First Look at the Data
- **Default of Credit Card Clients (PD Modelling)**
  - Data, Research Question and Method
  - First Look at the Data
- **Outlook and Potential Challenges**
- **References** (sources, data periods and licenses)

Stock Return Analysis: Data, Question and Method

Fama-French Factors and Asset Returns

Core Variables:

- **Stock Returns (R_i):** daily returns of 99 US-listed stocks (Yahoo Finance)
- **Risk-free Rate (R_f):** U.S. 1-Month T-Bill
- **Market Excess Return ($R_m - R_f$)**
- **Size Factor (SMB):** Small Minus Big
- **Value Factor (HML):** High Minus Low

Target:

- Excess equity returns ($R_i - R_f$)

Research Question:

How much of the cross-sectional return variation in a selected portfolio consisting of 100 selected stocks can be explained by factor premia?

Method:

OLS Regression and robust regression techniques applied to Fama-French type models

Method: the Fama–French Three-Factor Model

Fama-French Stock Return Modelling

Fitted separately for each stock on daily returns:

$$R_{i,t} - RF_t = \alpha_i + \beta_i \,(\text{Mkt-RF})_t + s_i \,\text{SMB}_t + h_i \,\text{HML}_t + \varepsilon_{i,t}$$

α_i: return the factors leave unexplained. R^2: share of return variance they explain.

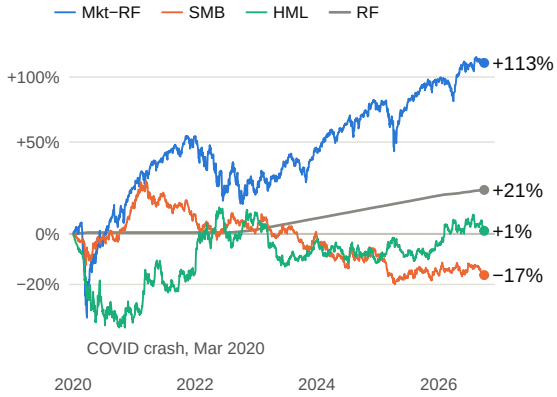
- **Mkt-RF, market premium:** value-weighted return of all US-listed stocks minus RF. β_i: the stock's market exposure.
- **SMB, small minus big:** small-cap minus large-cap returns, the size premium. $s_i > 0$: moves like a small cap.
- **HML, high minus low book-to-market:** return of value minus growth stocks. $h_i > 0$: value tilt, $h_i < 0$: growth tilt.
- **RF, risk-free rate:** one-month US Treasury bill. Subtracting it gives excess returns.

First Look at the Data

Fama-French Factors and Asset Returns

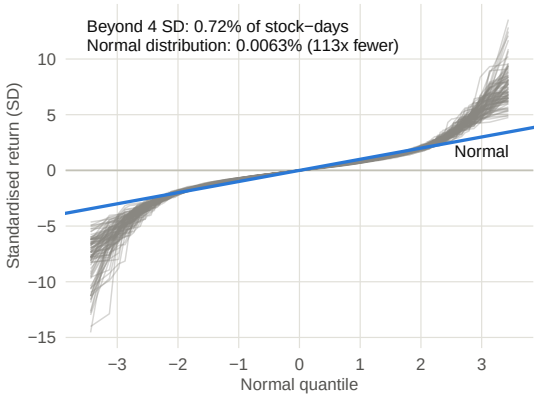
Market compounds, size and value swing

Cumulative factor returns, 2020 to Sep 2026, log scale



Daily returns have fat tails

Normal QQ plot, one grey line per stock (99)



Credit Card Defaults: Data, Question and Method

UCI Taiwan Credit Card Dataset

Features ($N = 30\,000$):

- **Demographics:** `LIMIT_BAL`, `SEX`, `EDUCATION`, `AGE`
- **History:** Repayment status (`PAY_0` to `PAY_6`)
- **Financials:** Bill statement amounts & previous payments

Target:

- `default.payment.next.month` (Binary, ~22% Defaults)

Research Question:

How to efficiently detect probability of default and model it in a non balances dataset?

Method:

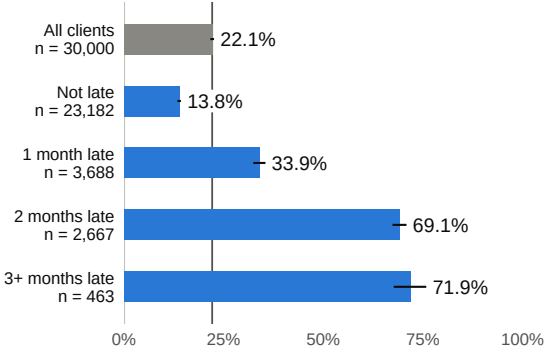
Probabilistic regression type models based on selected features

First Look at the Data

UCI Taiwan Credit Card Dataset

22% default; risk jumps once clients are late

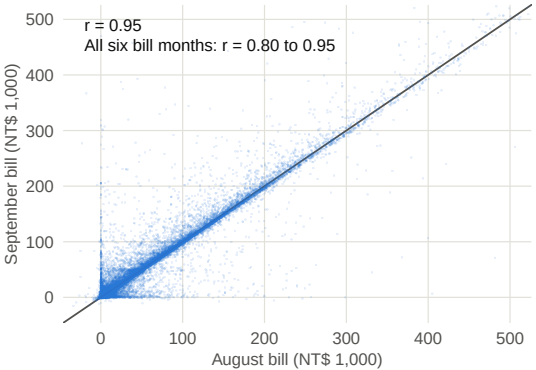
Share defaulting next month, by repayment status



Status in Sep 2005 (PAY_0); codes up to 0 count as not late.
Whiskers: 95% Wilson interval.

Monthly bill amounts are nearly collinear

Sep vs Aug bill, one dot per client; line: same amount



Axes cut at NT$ 500k; 60 of 30,000 clients lie outside.

Outlook and Potential Challenges

- **Stock Return Analysis:**
  - Non-stationarity and structural breaks in volatility regimes leading to potential outliars.
  - Non-normality of returns.
- **Credit Card Default Modelling:**
  - **Class Imbalance:** Models risk underpredicting rare defaults without re-weighting or threshold tuning.
  - Potential Multicollinearity in the data and selection of key factors for modelling.

References

Credit Card Default Dataset

- **Paper:** Yeh, I.-C., & Lien, C.-H. (2009). The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. *Expert Systems with Applications*, 36(2), 2473–2480. https://doi.org/10.1016/j.eswa.2007.12.020
- **Repository:** Yeh, I.-C. (2009). *Default of Credit Card Clients* [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C55S3H (CC BY 4.0). *Data period: April–September 2005.*

Fama-French Factor Data

- **Foundational Paper:** Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56. https://doi.org/10.1016/0304-405X(93)90023-5
- **Data Source:** French, K. R. (2026). *Data Library: Fama/French 3 Factors [Daily]*. Tuck School of Business, Dartmouth College. http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html *Data used: 2 Jan 2020 – 30 Sep 2026. Freely available; no formal license is stated.*

Financial Asset Pricing Data

- **Market Data:** Yahoo! Finance. (2026). *Historical Equity Price Data*. https://finance.yahoo.com/ (Fetched via R-Paket `quantmod`). *Data used: 2 Jan 2020 – 29 Oct 2026. Academic use under Yahoo's terms.*